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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/777,548	05/17/2022	Satoshi HATADA	SHX3C3	6668

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EXAMINER
CARLSON, MARC

ART UNIT	PAPER NUMBER
3723	

NOTIFICATION DATE	DELIVERY MODE
11/14/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/777,548	Applicant(s) HATADA et al.	
	Examiner MARC CARLSON	Art Unit 3723	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 8/22/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2 and 4-10 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2 and 4-10 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 8/22/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

1. The following is a Final Office Action on the merits.

Response to Amendment

2. Acknowledgement is made to the amendment received August 22, 2024, amending Claim 1 and adding new claims 8-10. Claim 3 was cancelled.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

> Determining the scope and contents of the prior art.
> Ascertaining the differences between the prior art and the claims at issue.
> Resolving the level of ordinary skill in the pertinent art.
> Considering objective evidence present in the application indicating obviousness or nonobviousness.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any

correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

3. As necessitated by Amendment, Claim 1-10 are rejected under 35 U.S.C. 103 as being unpatentable over Inoue WO 00/49911 (hereafter Inoue) in view of Kanamaru WO 2017/155039 (hereafter Kanamaru) and in further view of Jeannet et al. US 5,564,152 (hereafter Jeannet et al.).

a. Regarding Amended Claim 1, Inoue teaches:

1. (Currently Amended) A toothbrush comprising:

a handle body (labeled in attached Figure 4 below) made of a hard resin (plastic resin); and

a brush molded body (labeled in attached Figure 4 below) made of a soft resin (rubber) and provided on a distal end side of the handle body (shown in attached Figure 4 below),

wherein the handle body has an insertion portion (labeled in attached Figure 5(2) below) provided to protrude to the distal end side (shown in attached Figure 5(1) below),

the brush molded body has a head base portion (labeled in attached Figure 5(1) below) and a filament (labeled in attached Figure 5(1) below) configured to protrude to a front surface side of the head base portion (shown in attached Figure 5(1) below),

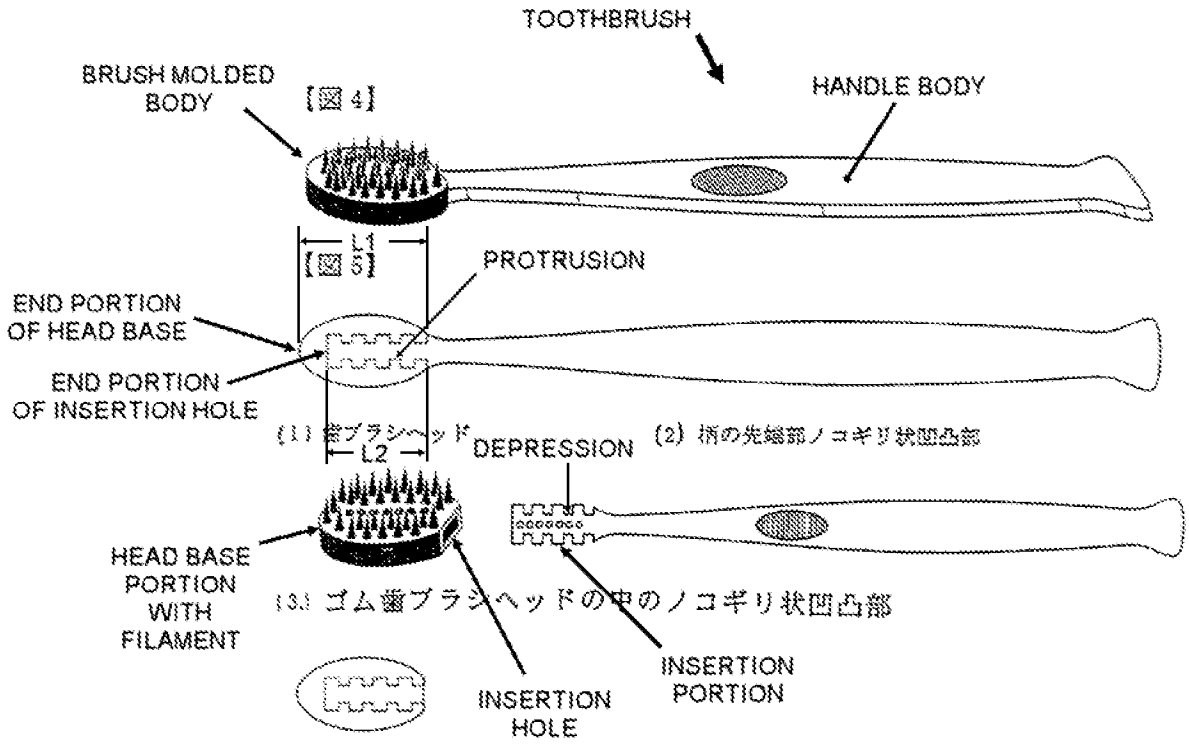
the head base portion has an insertion hole (labeled in attached Figure 5(1) below) into which the insertion portion is inserted and fitted (shown in attached Figures 4 and 5 below),

the brush molded body has a maximum distance between a base end of the filament and a back surface of the head base portion of 2.0 mm or more and 4.5 mm or less (see discussion below), and when a maximum length of the head base portion in an insertion direction is defined as L1 (labeled in attached Figure 5 below) and a maximum length of the insertion hole in the insertion direction is defined as L2 (labeled in attached Figure 5 below), a ratio of the maximum length of the insertion hole to the maximum length of the head base portion, which is represented by L2/L1, is 0.30 or more and 0.70 or less (see discussion below – 0.59 resulting value in Figure 2 below),

wherein the head base portion (labeled in attached Figure 5(1) below) has a thin-wall region (note that the size of the region can be arbitrarily identified due to undefined boundary) in which the distal end side of the back surface is gradually inclined to the front surface side to be gradually thinner toward the distal end side (see discussion below, shown in Figure 2 below),

wherein a maximum length of the thin-wall region in the insertion direction is 40% or more of the maximum length of the head base portion in the insertion direction (see discussion below – 84% resulting value in Figure 2 below) and

wherein a minimum distance(S) between the base end of the filament positioned on the most distal end side and the back surface of the head base portion is 1.5mm to 3.0 mm (see discussion below - 2.44mm resulting value in Figure 2 below).



Inoue discloses substantially all the limitations of the claim(s) except for providing a maximum distance between a base end of the filament and back surface of the head base portion as claimed. As presented, the claimed dimension is equivalently the thickness of the brush head. The Inoue device appears to be a standard size toothbrush, and therefore, sized to typical dimensions. The reference Kanamaru discloses a standard size toothbrush 1 shown in Figures 1 and 2. Kanamaru discloses t thickness “t” labeled in Figure 2 is 2.0-4.0mm. It would have been obvious to one having ordinary skill before the effective filing date of the claimed invention to size the thickness of the Inoue to be consistent with the thickness of common prior art toothbrushes as taught by Kanamaru. Therefore, it would have been obvious that the maximum distance

between the base end of the filament and back surface of the head base portion as claimed would be 2.0-4.0mm which is within the claimed range.

Using Figure 5 of Inoue, the claimed ratio calculates to roughly 0.78. Inoue discloses the use of a rubber material for the head base portion which can range from a very soft material to a firmer material. It would have been obvious to one having ordinary skill before the effective filing date of the claimed invention that the depth of the insertion hole would be dependent on the need to support and connect the rubbery head base portion. It is common engineering knowledge that a softer material would need a longer insertion distance to provide mechanical support for the head whereas a firmer material would need less insertion depth since the material is self-supporting. Additionally, a softer material would need more aggressive interlocking between the mating parts to prevent deformation during normal brushing operation from releasing the parts, where a firmer material would need a less aggressive interlocking feature to allowing the insertion without excessive force and the firm material would not be prone to deflecting (and releasing the parts) during normal operation. Therefore, sizing the depth of the insertion hole relative to the overall length would have been obvious based on the material properties of the soft resin selected for the brush molded body, therefore the claimed ratio 0.30 or more and 0.70 or less, which covers the middle of the overall range, would have been obvious since it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. *In re Boesch*, 617 F.2d 272, 205 USPQ 215 (CCPA 1980).

Inoue in combination of Kanamaru disclose a head shape the is fairly uniform in thickness from the handle attachment (front) to the distal end. Therefore, the combination of Claim 1 does not teach the thin-wall region as claimed. The reference Jeannet et al. discloses a head shape that as shown in Figures 2 and 3 that has a thin-wall region in which the distal end side of the back surface is gradually inclined to the front surface side to be gradually thinner toward the distal end side as claimed. That being said, since the thin-wall region does not have a claimed or inherent boundary, the region can be arbitrarily selected such that the thin-wall region in the insertion direction is 40% or more of the maximum length of the head base portion in the insertion direction as claimed. It would have been obvious to one having ordinary skill before the effective filing date of the claimed invention to modify the shape of the Inoue in view of Kanamaru combined device of Claim 1 to have a shape as taught by Jeannet et al. which results in the claimed thin-wall region with the motivation to have a different cosmetic shape that appears more streamlined.

The Applicant claims a lot of dimensions of the device. To provide additional insight, the Examiner has created new figures where he duplicated the Inoue device, modified the scale to match the dimensions of Kanamaru, and modified the shape to match the taper angle of the Jeannet et al. device. The goal of the exercise is to demonstrate using actual features and dimensional values from the prior art how one with ordinary skill in the art at the time of the invention would combine the teachings and by using actual dimensions be able to approximately measure the resulting claimed features as a sanity check of part

viability. The Figure 1 shows the insertion portion matching the ratio taught in Inoue. This would be viable for a scenario where the brush molded body were made of a material soft enough to deflect to allow the insertion portion to be installed.

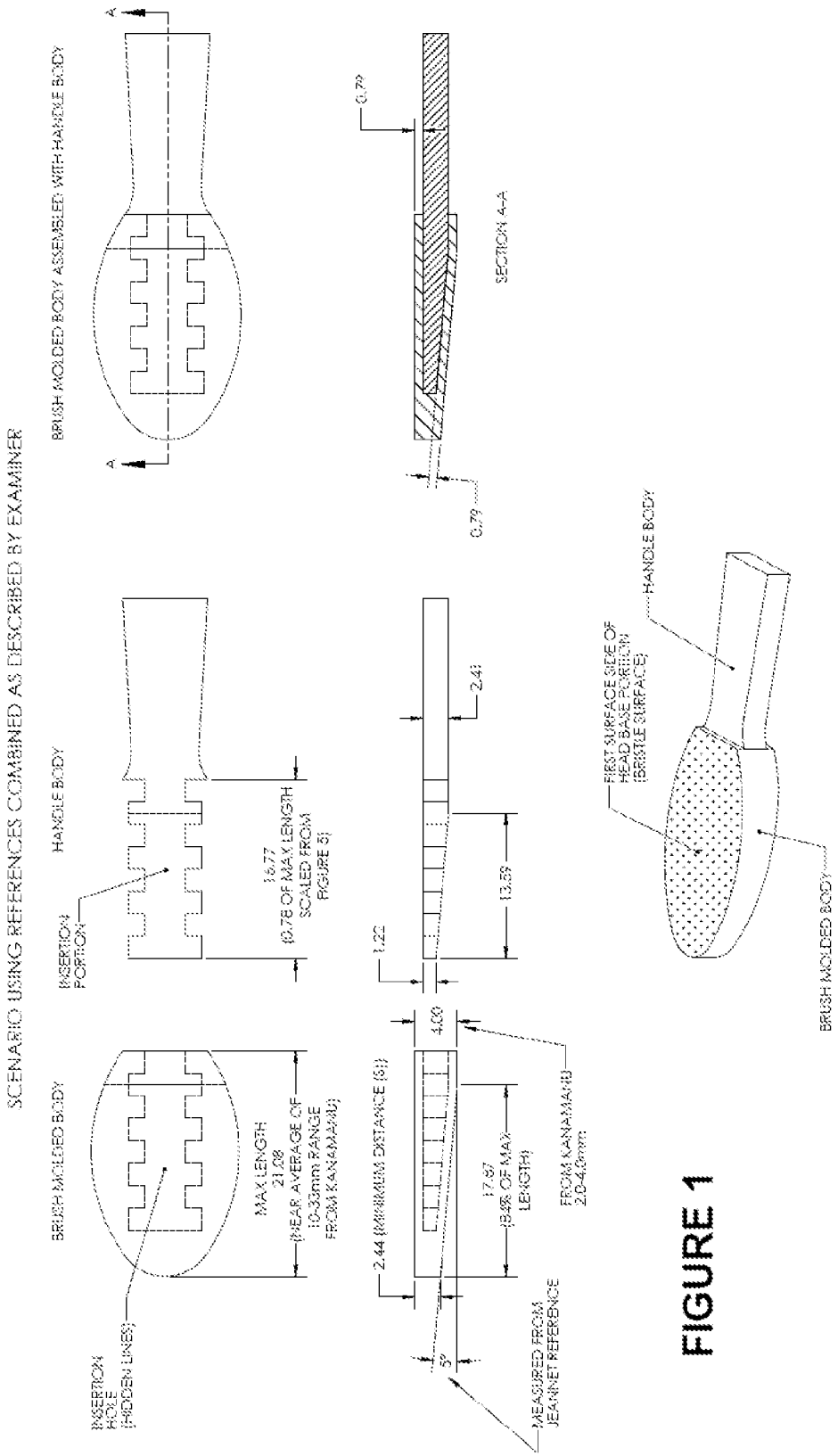

SCENARIO USING REFERENCES COMBINED AS DESCRIBED BY EXAMINER
BRUSH MOLDED BODY ASSEMBLED WITH HANDLE BODY
A
A
SECTION A-A
0.79
0.79
BRUSH MOLDED BODY
HANDLE BODY
INSERTION PORTION
16.77
(0.78 OF MAX LENGTH SCALED FROM FIGURE 5)
BRUSH MOLDED BODY
INSERTION HOLE (HIDDEN LINES)
MAX LENGTH 21.09
(NEAR AVERAGE OF 19-33mm RANGE FROM KANAMARU)
2.44 (MINIMUM DISTANCE (61))
4.00
1.22
2.41
13.59
17.87
(84% OF MAX LENGTH)
5°
MEASURED FROM JEANNET REFERENCE
FROM KANAMARU 20.4-29mm
FIRST SURFACE SIDE OF HEAD BASE PORTION (BRISTLE SURFACE)
HANDLE BODY
BRUSH MOLDED BODY

FIGURE 1

In a scenario where a firmer brush tip is desired, it would be obvious to make the material of a stiffer formulation. That being said, the stiffer material would obviously make it more difficult to get the insertion portion fully inserted. Therefore, it would have been obvious to make the insertion portion, and therefore the insertion hole, shorter to improve the likelihood of full insertion. As previously presented, it would have been obvious since it has been held that discovering an optimum value of a result effective variable, such as the insertion length based on the desired material stiffness, involves only routine skill in the art. *In re Boesch*, 617 F.2d 272, 205 USPQ 215 (CCPA 1980). In Figure 2 below the Examiner has shortened the Inoue device by eliminating one tooth in preparation for a stiffer material. The resulting modification results in a ratio of 0.59 which ends up being within the claimed range. This would be viable for a scenario where the brush molded body were made of a stiffer material that would not deflect enough to allow the longer insertion portion. Note that the Examiner's Figure 2 design combining the prior art references to scale and obvious knowledge of material elastic properties results in sizes and wall thicknesses that would produce a viable product. It is important to point out, that obvious experimentation with actual device materials would allow the exact optimal dimensions to be determined. Figure 2 will be used to as a reference to teach claimed values resulting from the obvious combination of prior art and obvious design knowledge at the time of filing.

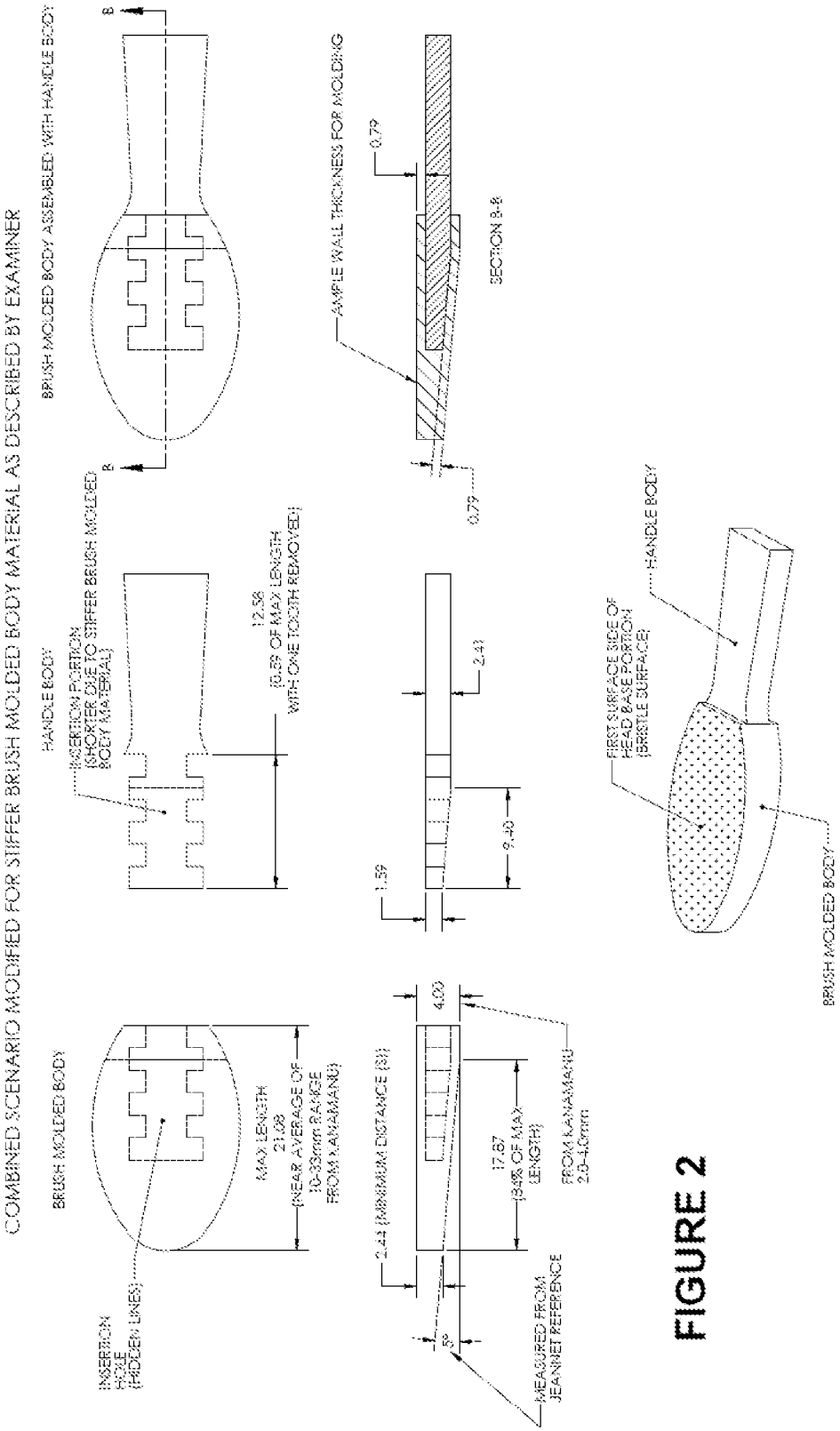

COMBINED SCENARIO MODIFIED FOR STIFFER BRUSH MOLDED BODY MATERIAL AS DESCRIBED BY EXAMINER
BRUSH MOLDED BODY ASSEMBLED WITH HANDLE BODY
HANDLE BODY
INSERTION PORTION (SHORTER DUE TO STIFFER BRUSH MOLDED BODY MATERIAL)
12.58 (0.59 OF MAX LENGTH WITH ONE TOOTH REMOVED)
AMPLE WALL THICKNESS FOR MOLDING
0.79
SECTION B-B
BRUSH MOLDED BODY
INSERTION HOLE (HIDDEN LINES)
MAX LENGTH 21.08 (NEAR AVERAGE OF 10-33mm RANGE FROM KANAMANU)
2.44 (MINIMUM DISTANCE (B))
4.00
17.87 (84% OF MAX LENGTH)
FROM KANAMANU 2.0-4.0mm
MEASURED FROM JEANNET REFERENCE
1.59
2.41
9.40
FIRST SURFACE SIDE OF HEAD BASE PORTION (BRISTLE SURFACE)
HANDLE BODY
BRUSH MOLDED BODY

FIGURE 2

b. Regarding Claim 2, Inoue teaches:

2. (Original) The toothbrush according to Claim 1,

wherein a distance between an end portion of the insertion hole (labeled in attached Figure 5) above) on the distal end side and an end portion of the head base portion (labeled in attached Figure 5) above) on the distal end side, which is represented by L1 - L2, is 5 mm or more and 15 mm or less (see discussion below – 8.5mm resulting value in Figure 2 above).

Inoue discloses substantially all the limitations of the claim(s) except for providing a distance between an end portion of the insertion hole and an end portion of the head base portion as claimed. As presented, the claimed dimension is equivalently the thickness of the brush head. As previously presented, it would have been obvious to depend upon the Kanamaru reference for providing drawings for the features taught by Inoue. That being said, Kanamaru discloses that the head length (claimed L1) is set in the range of 10mm to 33mm. That being said, using the previous calculation of a 0.78 ratio from Inoue, the cited distance would be 2.2mm to 7.26mm. Therefore, as previously presented in Claim 1, it would have been obvious to size the insertion features to achieve a desired attachment based on the material properties of the soft resin. Therefore, the claimed distance is an obvious result of this optimization and the need to have enough material thickness at the distal end to provide sufficient material to provide the mechanical support.

c. Regarding Claim 4, Inoue teaches:

4. (Currently amended) The toothbrush according to ~~any one of~~ Claim[[s]] 1~~ to 3~~,

wherein the insertion portion (labeled in attached Figure 5(2) above) has at least one or more depressions (labeled in attached Figure 5(2) above), which are inwardly concave, at an edge in a direction orthogonal to a protrusion direction of the filament (shown in attached Figures 5(1) and 5(2)), and

the head base portion (labeled in attached Figure 5(1) above) has a protrusion (labeled in attached Figure 5 above) at a position at which the head base portion is fitted into the depression when the insertion portion is inserted into the insertion hole (shown in Figures 5 and 5(1)).

d. Regarding Claim 5, Inoue teaches:

5. (Original) The toothbrush according to Claim 4,

wherein the depression (labeled in attached Figure 5(2) above) is provided at an edge in a width direction orthogonal to the protrusion direction of the filament and the insertion direction (shown in attached Figures 5(1) and 5(2)), and is positioned on a rear end side with a distal end of the insertion portion in the insertion direction as an origin in a range of 30% or more and 70% or less of the maximum length of the insertion portion in the insertion direction (labeled depression is within the claimed range as shown in Figure 5(2) above).

e. Regarding Claim 6, Inoue teaches:

6. (Currently amended) The toothbrush according to ~~any one of~~ Claim[[Es]] 1~~ to 5~~,

wherein the soft resin is polyurethane.

Inoue discloses substantially all the limitations of the claim(s) except for disclosing the soft resin is rubber rather than polyurethane. It would have been obvious to one having ordinary skill before the effective filing date of the claimed invention to elect a polyurethane material which is commonly available in formulations with similar mechanical properties to the rubber originally disclosed by Inoue with the motivation to improve the color options and moldability, since it has been held to be within the general skill of a worker in the art to select a known material on the basis of its suitability for the intended use as a matter of obvious design choice. *In re Leshin*, 125 USPQ 416. See also *Ballas Liquidating Co. v. Allied industries of Kansas, Inc.* (DC Kans) 205 USPQ 331.

f. Regarding Claim 7, Inoue teaches:

7. (Currently amended) The toothbrush according to ~~any one of~~ Claim[[Es]] 1~~ to 6~~, wherein, with respect to a protrusion direction of the filament (shown in Figure 5(1), a position of the base end of the filament (bristle surface as shown in Figure 5(1)) is the same as a position of a parting surface of the brush molded body (see discussion below).

As shown in Figure 5(1) above, Inoue discloses what appears to be a molded rubber part with an insertion hole as labeled. It would have been obvious design choice to one having ordinary skill before the effective filing date of the claimed invention that the part is formed using a top and bottom molding tool with the insertion hole core movable sideways from the cavity. That being said, the

decision to place the parting line coincident with the topmost surface shown would be obvious for the loading and embedding of bristles into one side of the molding tool and injecting the material into the cavity to embed the bristles with the overall shape shown in Figures 5(1), 5(4), and 5(5).

g. Regarding New Claim 8, Inoue teaches:

8. (New) The toothbrush according to 1, a minimum thickness of the head base portion (labeled in attached Figure 5(1) above) is 1.0 to 2.5 mm (see discussion below – 2.44mm resulting value in Figure 2 above)

As shown in Figure 2, the thickness is a result of the desired angle and desired length of the thin wall region, with a minimum value based on material properties. It would have been obvious since it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. *In re Boesch*, 617 F.2d 272, 205 USPQ 215 (CCPA 1980).

h. Regarding New Claim 9, Inoue teaches:

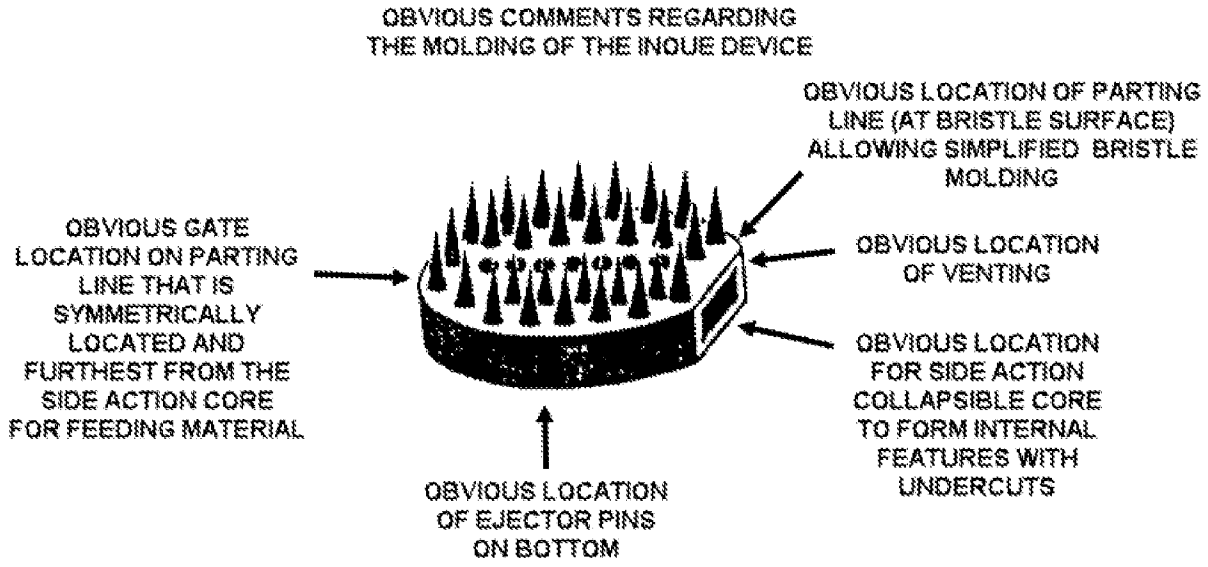
9. (New) The toothbrush according to 1, a thickness of the head base portion (labeled in attached Figure 5(1) above) between an upper surface (bristle surface) of the front side of the head base portion and a surface of the insertion hole (labeled in attached Figure 5(1) above) on the upper surface side is 0.5 mm to 2.0 mm (see discussion below – 0.79mm resulting value in Figure 2 above).

As shown in Figure 2, the thickness is a result of a need to size the overall thickness of the insertion portion and the thickness of the head base portion to achieve the mechanical strength yet maintain a minimum value based on material properties and moldability of the soft material. It would have been obvious since it has been held that discovering an optimum value of a result effective variable involves only routine skill in the art. *In re Boesch*, 617 F.2d 272, 205 USPQ 215 (CCPA 1980).

i. Regarding New Claim 10, Inoue teaches:

10. (New) The toothbrush according to 1, the head base portion (labeled in attached Figure 5(1) above) has a gate mark at the distal end of the head base portion (see discussion below).

The location of the gate for would be determined by a mold tooling designer with ordinary skill in the art at the time of filing. Inoue discloses that the head base portion is molded. Therefore, locating the gate would be an obvious step in the mold design process. Based on solely on experience with designing mold tooling prior to filing, it would be common knowledge to build the Inoue part using the following obvious tooling details. Trimming the gate off the part after molding would obviously result in the claimed gate mark at the gate/part location.



Response to Arguments

4. Rejections Under 35 U.S.C. 103

Applicant's arguments with amendments, filed August 22, 2024, with respect to the 35 U.S.C. 103 rejection(s) of Claims 1, 2, and 4-7 under Inoue WO 00/49911 in view of Kanamaru WO 2017/155039 and Claim 3 under Inoue WO 00/49911 in view of Kanamaru WO 2017/155039 in further view of Jeannet et al. US 5,564,152 have been fully considered and are not persuasive. However, as necessitated by amendment, Claims 1-10 are now rejected under Inoue WO 00/49911 in view of Kanamaru WO 2017/155039 in further view of Jeannet et al. US 5,564,152.

The Applicant provided detailed arguments regarding the molding, dimensions, and material thicknesses that he argues makes the combination of

the Inoue design with dimensions provided by Kanamaru used by the Examiner incompatible with tapering taught by Jeannet. The Applicant neglected to consider that his identified issues are resolved by the obvious design step of additionally tapering the insertion portion of the handle body to match the modification of the taper to the external surface of the brush molded body taught by Jeannet resulting in a uniform wall thickness. The Examiner has provided new Figure 1 that depicts a design resulting from the combination of prior art references to one with ordinary skill in the art at the time of filing. The Examiner has provided new Figure 2 that depicts an obvious modification in the Figure 1 design that would be necessary if a stiffer material were selected to shorten the length of the insertion portion to allow it to be fully seated. Although, obvious experimentation with the desired materials would be necessary to fully optimize the dimensions, the Examiner has disclosed a viable scenario that shows how one with ordinary skill would modify the length of the insertion portion and the impact to the part dimensions. Using dimensions from Kanamaru and an angle from Jeannet, the Examiner shows how obvious modifications to Inoue depicted in Figure 2 naturally results in the Applicant's claimed values when a stiffer material is desired.

Conclusion

5. **THIS ACTION IS MADE FINAL.** Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

6. Any inquiry concerning this communication or earlier communications from the examiner should be directed to MARC CARLSON whose telephone number is (571)272-9963. The examiner can normally be reached Monday-Thursday 6:30am-3:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, BRIAN KELLER can be reached on (571) 272-8548. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center,

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/MARC CARLSON/
Primary Examiner, Art Unit 3723